

Cosmos_C3

Getting Started

For HEX20

Focus project: UAE_FlatSat

23/Aug/2023

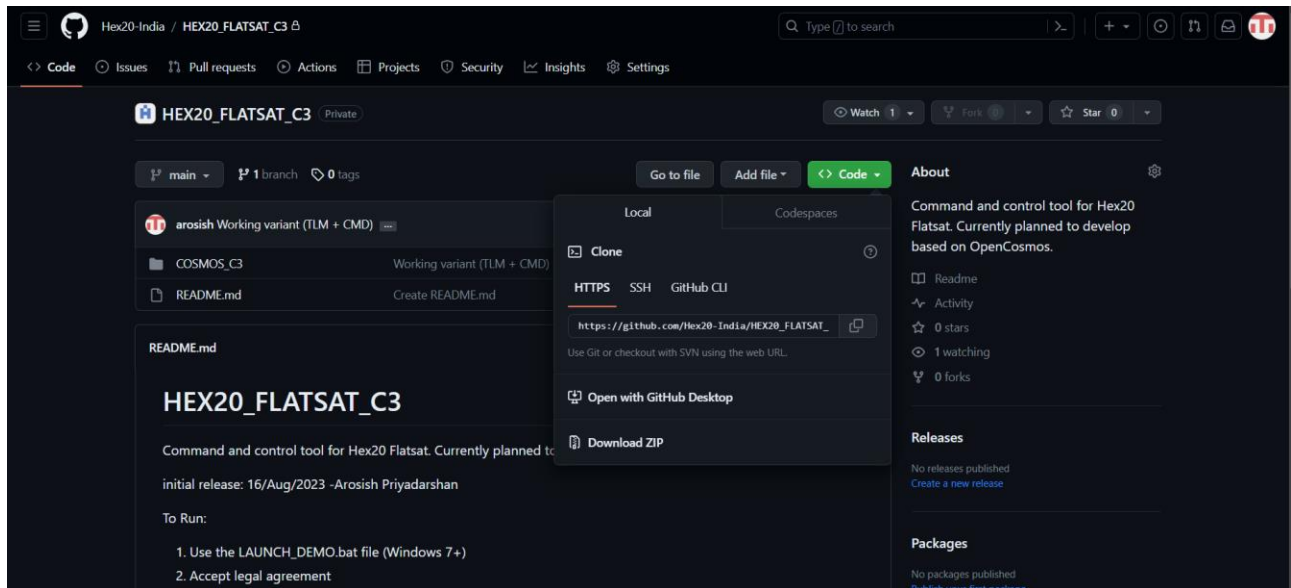
Arosish Priyadarshan

Initial Release

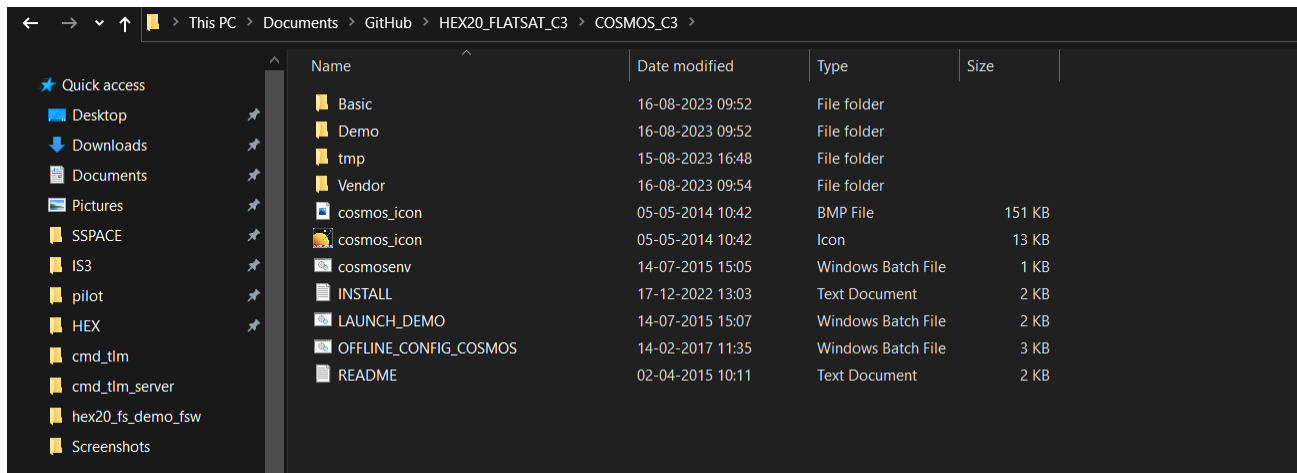
Install: (Windows 7+)

1. Download the repository from Github:

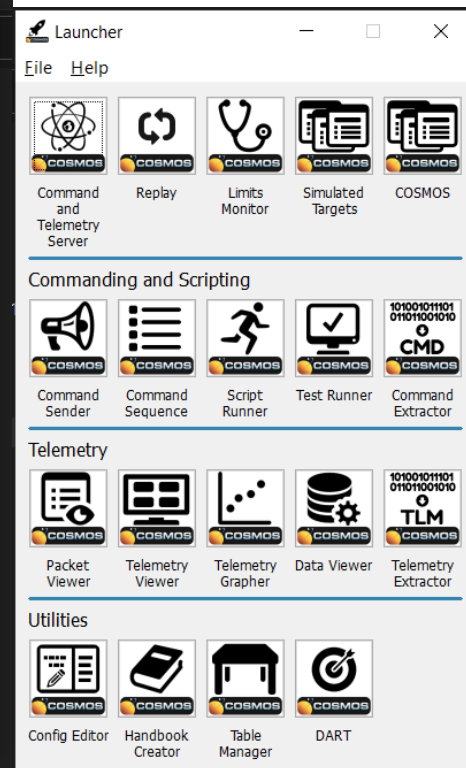
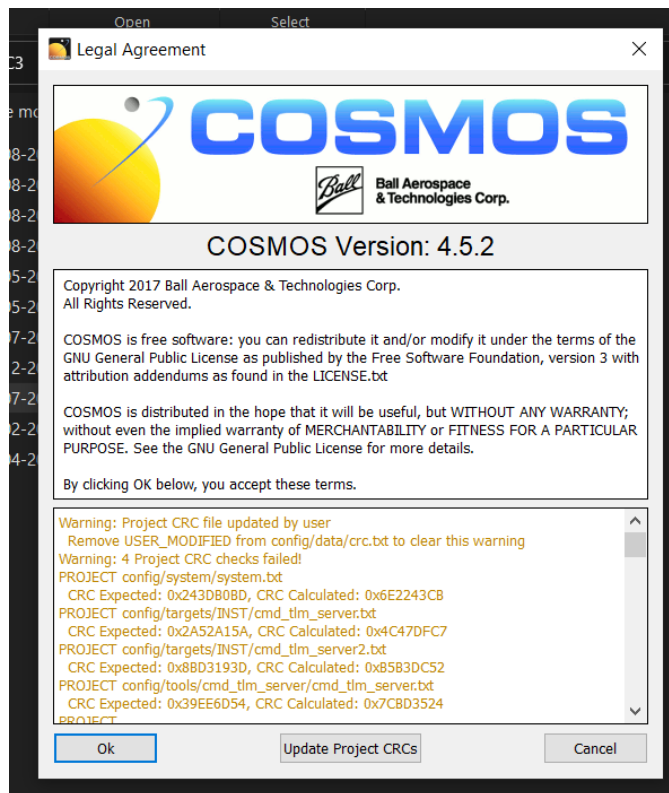
https://github.com/Hex20-India/HEX20_FLATSAT_C3.git



2. There is no need to do any installation, the unzipped file is sufficient to run the program. You should get all these files in the unzipped folder.



3. Run LAUNCH_DEMO to open the program

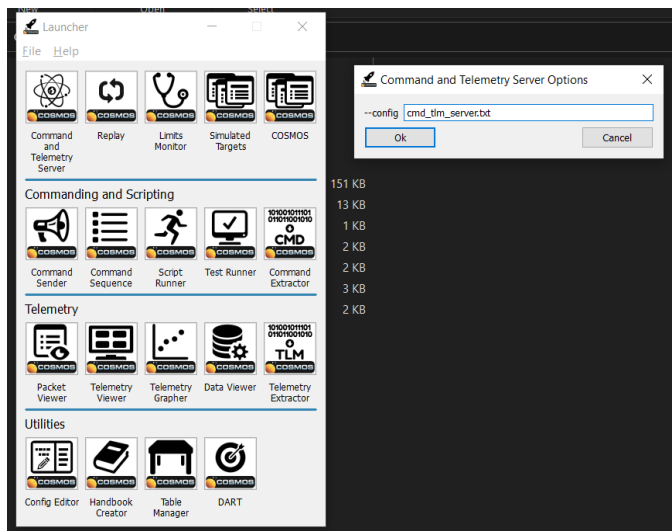


Use:

1. Run LAUNCH_DEMO to open the program
2. The major utilities we are going to use in the beginning are command and telemetry server, packet viewer and command sender.

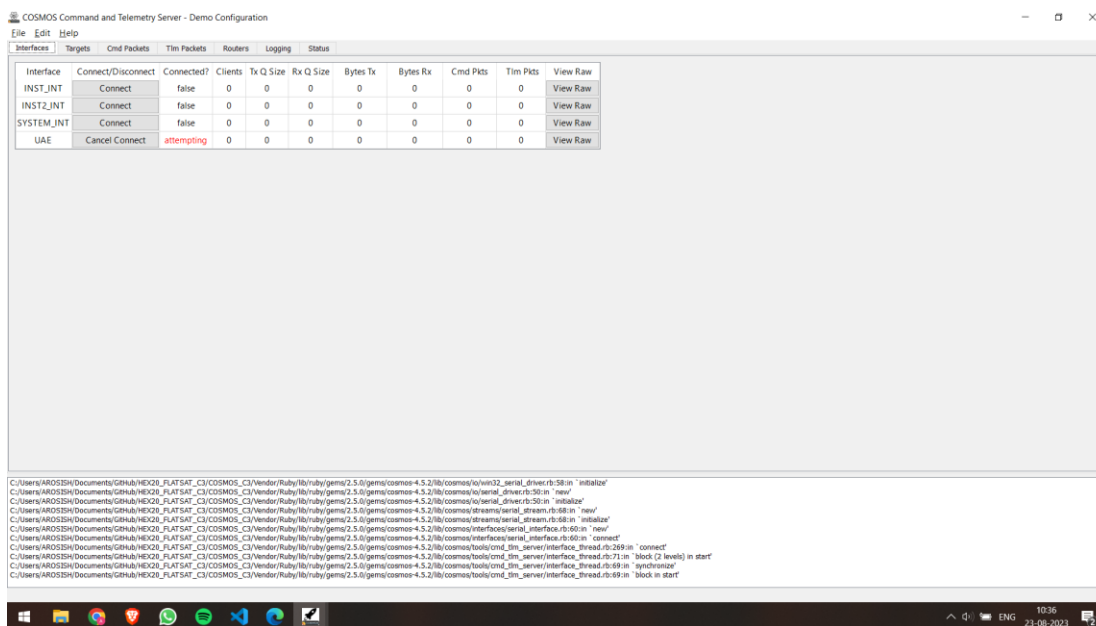
A) Command and telemetry server:

- a. Open the utility from the launcher application.



click ok, (note: the correct config file should be selected, i.e., "cmd_tlm_server.txt")

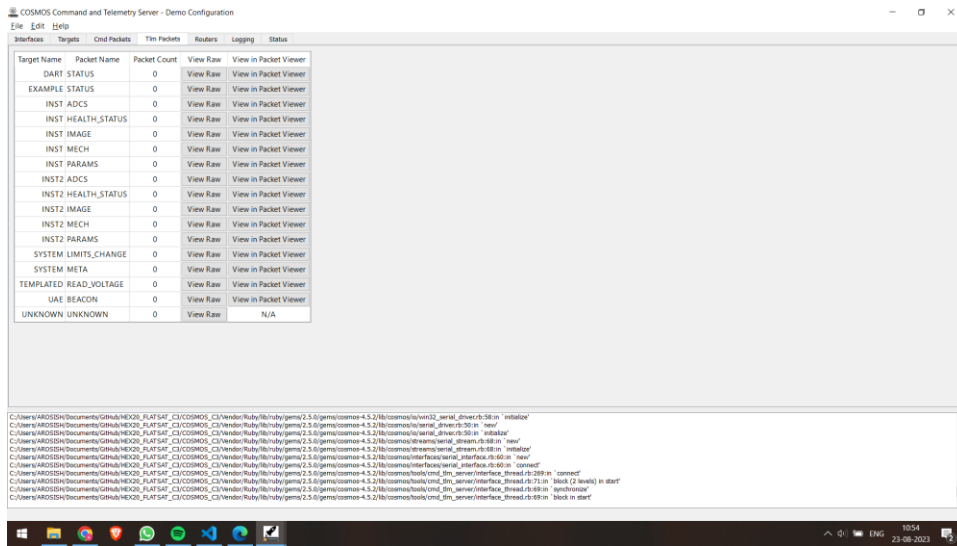
- b. The server will open up:



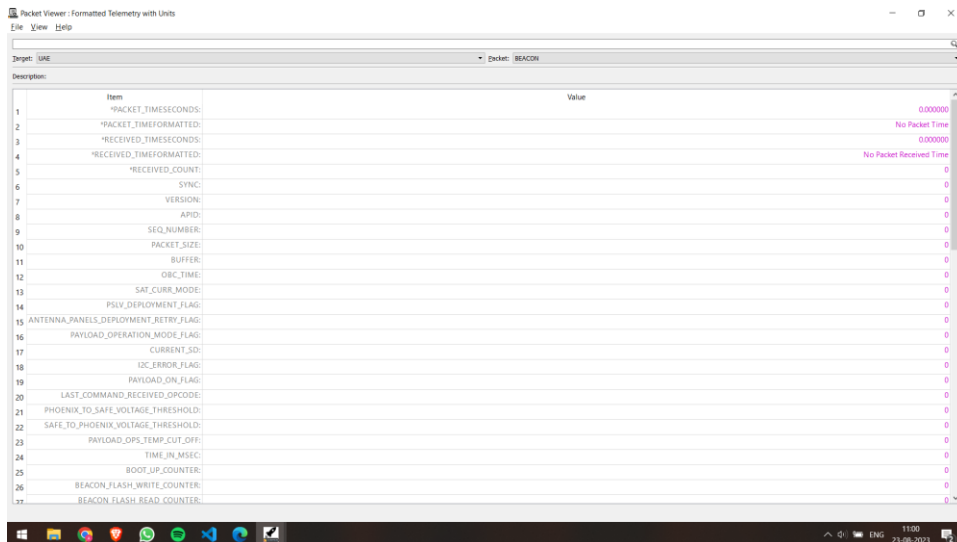
- c. We can try to connect to UAE or any other client (connection are defined by the config file, i.e.,
“...\GitHub\HEX20_FLATSAT_C3\COSMOS_C3\Demo\config\tools\cmd_tlm_server\cmd_tlm_server.txt” file)
- d. After connection, we can view the raw received and sent data from the server.
- e. We can also open the packet viewer (to get the decoded data) or command sender (to transmit command) via the server or through the launcher.

B) Packet viewer

- a. Go to the TLM packet, and open packet viewer for the required packet

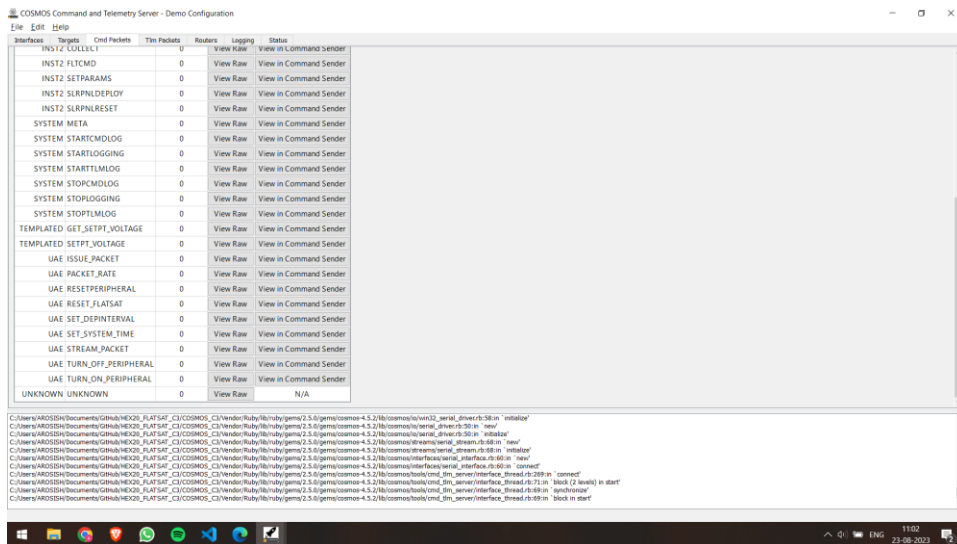


- b. Example packet: UAE Beacon

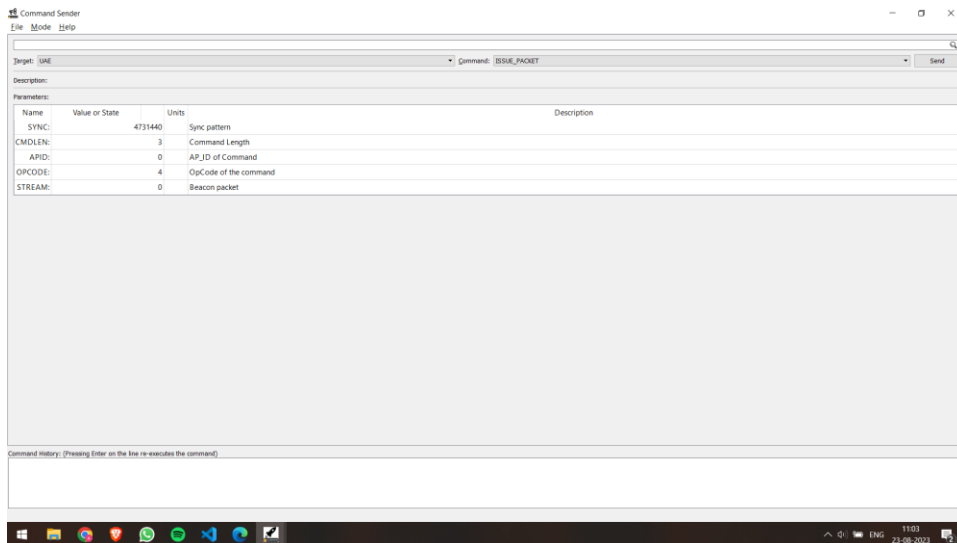


C) Command Sender:

a. Go to cmd Packets



b. Open Command Sender:



- c. We can switch between the commands and update the parameters (within the limits)
- d. Hit send to transmit the command.

Configuration:

The major configuration involves:

- a. Adding the target client
- b. Defining the connection
- c. Adding the telemetry packets
- d. Adding the command
- e. Further limits, scripts, telemetry screens and plots can be added

A) Adding Target:

- a. Go to the location :
“...\GitHub\HEX20_FLATSAT_C3\COSMOS_C3\Demo\config\system”.
- b. Open: “system.txt”
- c. And add the line:

DECLARE_TARGET UAE

i.e. this will define the target UAE for system

- d. Ref:

<https://ballaerospace.github.io/cosmos-website/docs/v4/system>

B) Defining the connection:

- a. Go to the location:
“...\GitHub\HEX20_FLATSAT_C3\COSMOS_C3\Demo\config\tools\cmd_tlm_server”
- b. Open: “cmd_tlm_server.txt”
- c. Add the lines:

INTERFACE UAE serial_interface.rb COM5 COM5 115200 NONE 1 5 nil FIXED 158

TARGET UAE

- d. Here a serial connection is defined on COM5 with baud of 115200, we can make different connection too, like TCP/IP or UDP. We have to add the specific interface config for those in the same file.
- e. Ref:

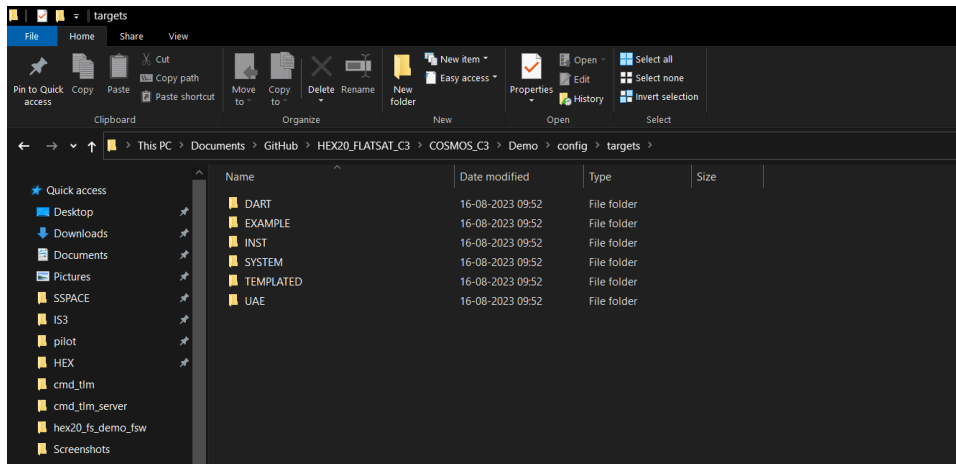
<https://ballaerospace.github.io/cosmos-website/docs/v4/interfaces>

C) Adding Telemetry and Commanding:

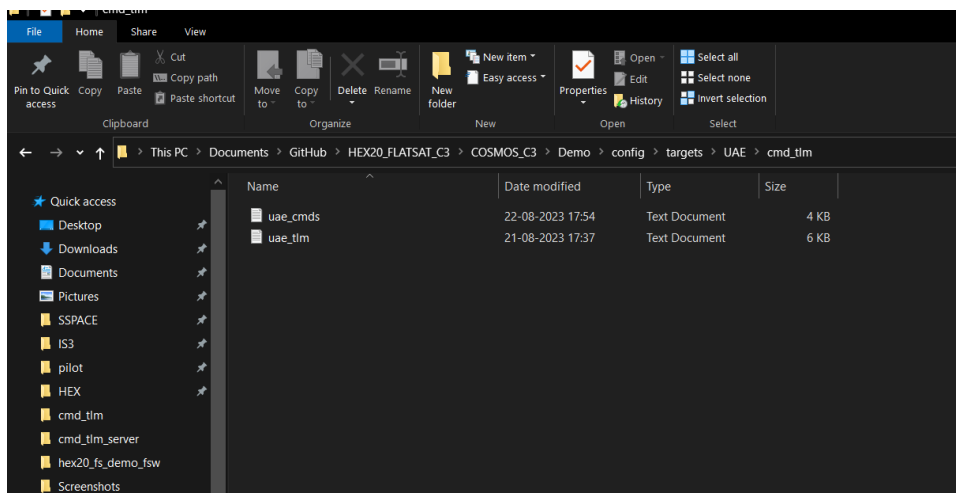
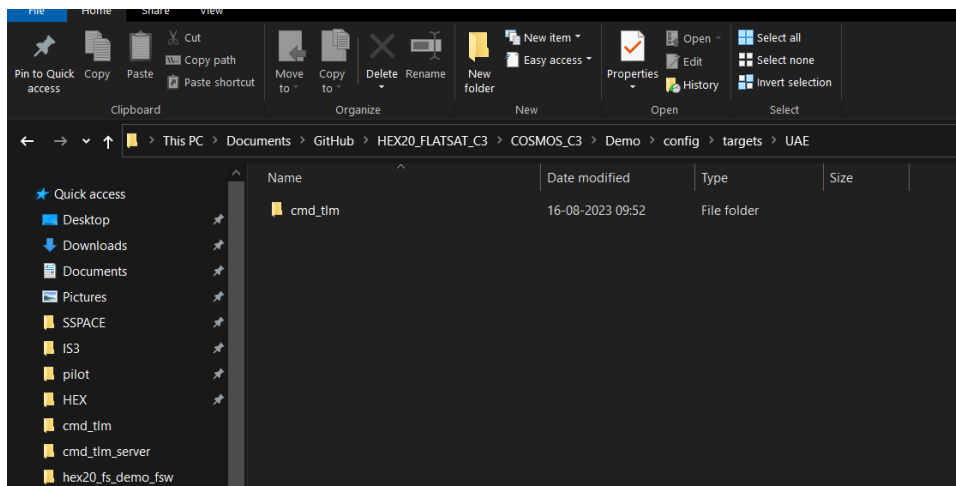
- a. Telemetry will be target specific.
- b. Make the target folder in the following location and name it in all CAPS:

“\GitHub\HEX20_FLATSAT_C3\COSMOS_C3\Demo\config\targets”

Folder: “UAE”



- You can add multiple targets, make sure the name is in all caps, and the target is defined in system file and connection interface is present in the cmd_tlm_server file.
- Add “cmd_tlm” folder in your target folder and add the cmd and tlm txt file in that folder.



- e. The command and telemetry file should be named as :

“uae_cmds” and “uae_tlm”

I.e., the name of target followed by file type

- f. The cmd and tlm file can be configured to add new telemetry packets, new parameters in these packets, update in the packet structure. Similar for commanding, we can change, fix and add or remove parameters or commands.
- g. Ref:

<https://ballaerospace.github.io/cosmos-website/docs/v4/command>

<https://ballaerospace.github.io/cosmos-website/docs/v4/telemetry>